

Philippe Decoste

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Robotist with industry experience building real-time vision systems. Published author at top robotics conference. Background in aerospace and current focus on applied mathematics for state estimation.

EDUCATION

Master of Applied Science in Mechanical Engineering

Sept. 2021 – Dec. 2023

McGill University, Montreal, Canada — DECAR Systems Group[§], Advisor: James Richard Forbes[§]

- Research in autonomous navigation and Bayesian state estimation in challenging environments using thermal cameras.
- Author on a paper[§] presented at the International Conference on Robotics and Automation (ICRA), the top conference in the field of robotics, for my contributions to a robust homography-based navigation method employing Lie groups.
- Built core expertise in optimization, state estimation, and differential geometry to derive Lie group-based Kalman filters and batch solvers for SLAM, incorporating covariance quantification for robust navigation in uncertain environments.
- Classes covered subjects such as model predictive control, machine/deep learning and optimization proofs; projects include building an optimal path planner in unknown environments and a numerical optimization library.

Bachelor of Engineering, Aerospace, Space Systems Design, Co-op Option

Sept. 2016 – April 2021

Carleton University, Ottawa, Canada

- Graduated with high distinction and obtained senate medal for high academic achievement.
- Design and analysis of aircraft, spacecraft, and rockets. Emphasis on performance & control theory of flights and orbits.
- Classes and projects included orbital mechanics, satellite design and control, communication links, trans-atmospheric propulsion.

WORK EXPERIENCE

Perception Systems Applied Scientist

February 2024 – August 2024 (Contract)

ARA Robotique, Montreal, Canada

- Engineered a state estimation framework for long-range (1.5+ km) dynamic object tracking, fusing monocular vision with GNSS data to provide uncertainty-informed position and velocity estimates, achieving <5% error in field tests.
- Integrated the algorithm into an aerial drone, using Lie theory to clearly and efficiently model dynamic frames of reference for real-time localization.
- Improved the project's documentation process by providing clear and concise explanations of complex mathematical concepts with a switch to LaTeX-based derivations and diagrams, leading to quicker onboarding of new team members.
- Developed the algorithm and its ROS integration in C++ / Python through collaboration on Git with proper CI/CD practices and unit/integration tested code achieving increased edge case coverage.

Spacecraft GNC Researcher

May 2020 – August 2020

Carleton University – Spacecraft Robotics and Control Laboratory, Ottawa, Canada

- Worked to develop and enhance the ASTRO multi-agent collision avoidance and trajectory optimization algorithm proposed by NASA astronaut Gregory Chamitoff under Professor Steve Ulrich.
- Investigated state of the art path-planning literature resulting in proposed improvements such as collaborative trajectory optimization protocols to enhance the solving efficiency of multi-UAV pathing and decrease computation time.
- Presented to industry professionals from Carleton University and NASA's Jet Propulsion Laboratory with great feedback.

Co-op Student in Aerostructures (Stress & Design) and Airworthiness

May 2019 – March 2020

Bombardier Aerospace / DeHavilland Aircraft of Canada Ltd. – Q400 Program, Toronto, Ontario

- Designed and 3D modeled structural components using CATIA, integrating them with existing aircraft structure to handle new loading requirements of aircraft modification project; interpreted relevant technical drawings.
- Led meetings with technical experts to assess impact on aircraft safety, reaching solutions for risk mitigation multilaterally across departments.

Space Science and Technology Intern

May 2018 – August 2018

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Canadian Space Agency – David Florida Laboratory (DFL) Satellite Testing Facility, Nepean, Ontario

- Authored a 70-page technical foresight report on lunar exploration and AIT infrastructure adaptation, incorporating design, thermal/EMC, and environmental considerations.
- Shadowed spacecraft testing campaigns including vibration and EMI/EMC qualification of CubeSats under the Canadian Satellite Design Challenge.
- Gained hands-on exposure to satellite hardware interfaces, environmental testing protocols, and the challenges of interplanetary system integration.

SKILLS

Software: ROS • OpenCV • OpenGL • Ansys STK • CATIA • Solid Works • Onshape • Linux • Docker • Git • LaTeX

Programming: Python • C++ • MATLAB • Simulink

Mathematical Tools: Probability and Statistics • Optimization • Computer Vision • SLAM • Control Theory • Path Planning • System Modelling and Simulation

PROJECTS

Hardware integration – Sensor Fusion Rig (Thesis) - *ROS, Docker, C++, Computer Vision*

- Integrated a thermal camera–IMU rig with ROS for real-world data collection and visual-inertial sensor fusion validation.
- Performed stereo camera-to-camera and camera-to-imu calibration to ensure sensor head accuracy.

Thermal-Inertial Navigation for GNSS-Denied Extreme Environments (Thesis) - *State estimation, computer vision, simulation, optimization, Python*

- Derived and prototyped a thermal signature-based visual-inertial navigation backend through numerical modeling, path-planning, and simulation in Python, enabling rapid, low-cost iteration prior to hardware deployment.
- Validated the thermal navigation system against motion capture ground truth, significantly outperforming conventional feature-based methods by leveraging a novel adaptation of a perception method from the field of control theory.

Towards Topographical SLAM[§] - *State estimation, computer vision, simulation, Python*

- 1st place winner of the Informs OR/MS poster competition at the graduate level for my developmental work towards ultralightweight robotic state estimation leveraging rangefinder measurements of local elevation instead of vision.

Information-Optimal Path Planning in Uncertain Environments – *Probability theory, path planning, simulation, SLAM*

- Reimplemented and added collision avoidance considerations to the iCR algorithm from the paper “Active Exploration and Mapping via Iterative Covariance Regulation over Continuous SE(3) Trajectories”.

Numerical Optimization Library – *Optimization, MATLAB*

- Built a MATLAB library implementing line search with Wolfe conditions, gradient descent, Newton and Quasi-Newton methods (e.g. BFGS).

Biomechanic Analysis Tool using Monocular Perception – *Computer vision*

- Integrated Google MediaPipe results with OpenCV to perform the calculation and display of limb and center of gravity positions to inform rock climbing success from a biomechanical analysis standpoint.

Convolutional Neural Network Image Classifier – *Machine/deep learning, optimization*

- Achieved 90% image classification success rate with original network architecture in a Kaggle competition for classifying low resolution images from the fashion MNIST dataset obfuscated by standard MNIST data.

INTERESTS

Skiing • Rock Climbing • Cycling • Soccer • Simulation development